RHWP Scan

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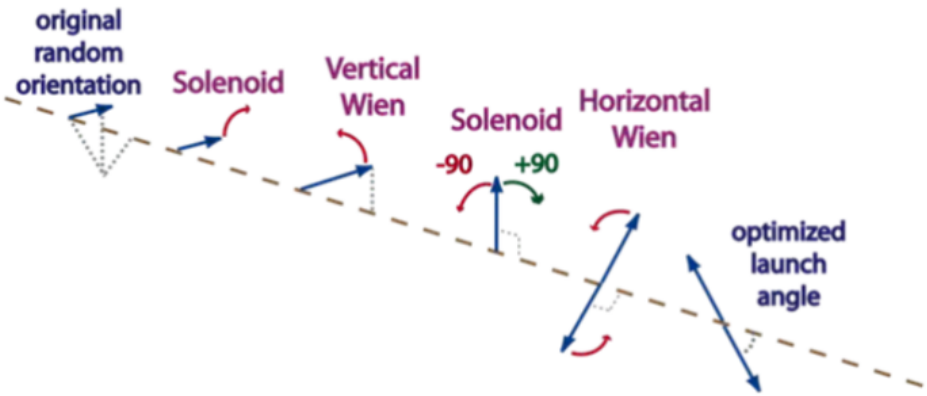
Overview

We have four Wien settings:

- 1) VWien $= 0.0^\circ$, no flip
- 2) VWien $= +90.0^\circ$, flip right
- 3) VWien $= -90.0^\circ$, flip right
- 4) VWien $= +90.0^\circ$, flip left



Goal: We want to do a PITA scan, position scan, and RHWP study for all four Wien settings.

This Study

Perform a PITA scan, position scan, and RHWP study for the 1st Wien settings

- 1) VWien $= 0.0^\circ$, no flip
- 2) VWien $= +90.0^\circ$, flip right
- 3) VWien $= -90.0^\circ$, flip right
- 4) VWien $= +90.0^\circ$, flip left

Helicity Board Configuration

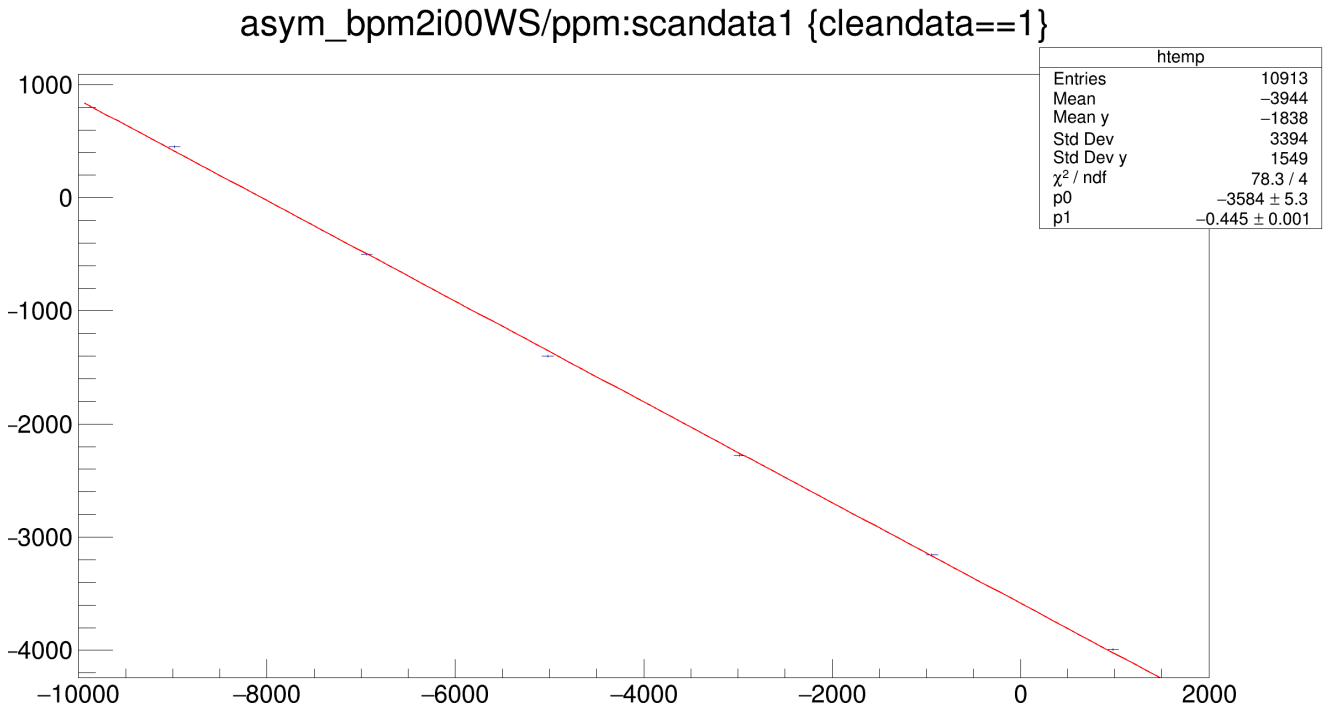
- Pattern: Octet
- Delay: 16
- Freq: 240 Hz
- T_{settle}: 100 μs
- T_{stable}: 4066.65 μs
- Current: 65μA

POLOG: `https://logbooks.jlab.org/entry/4473341`

PITA Scan

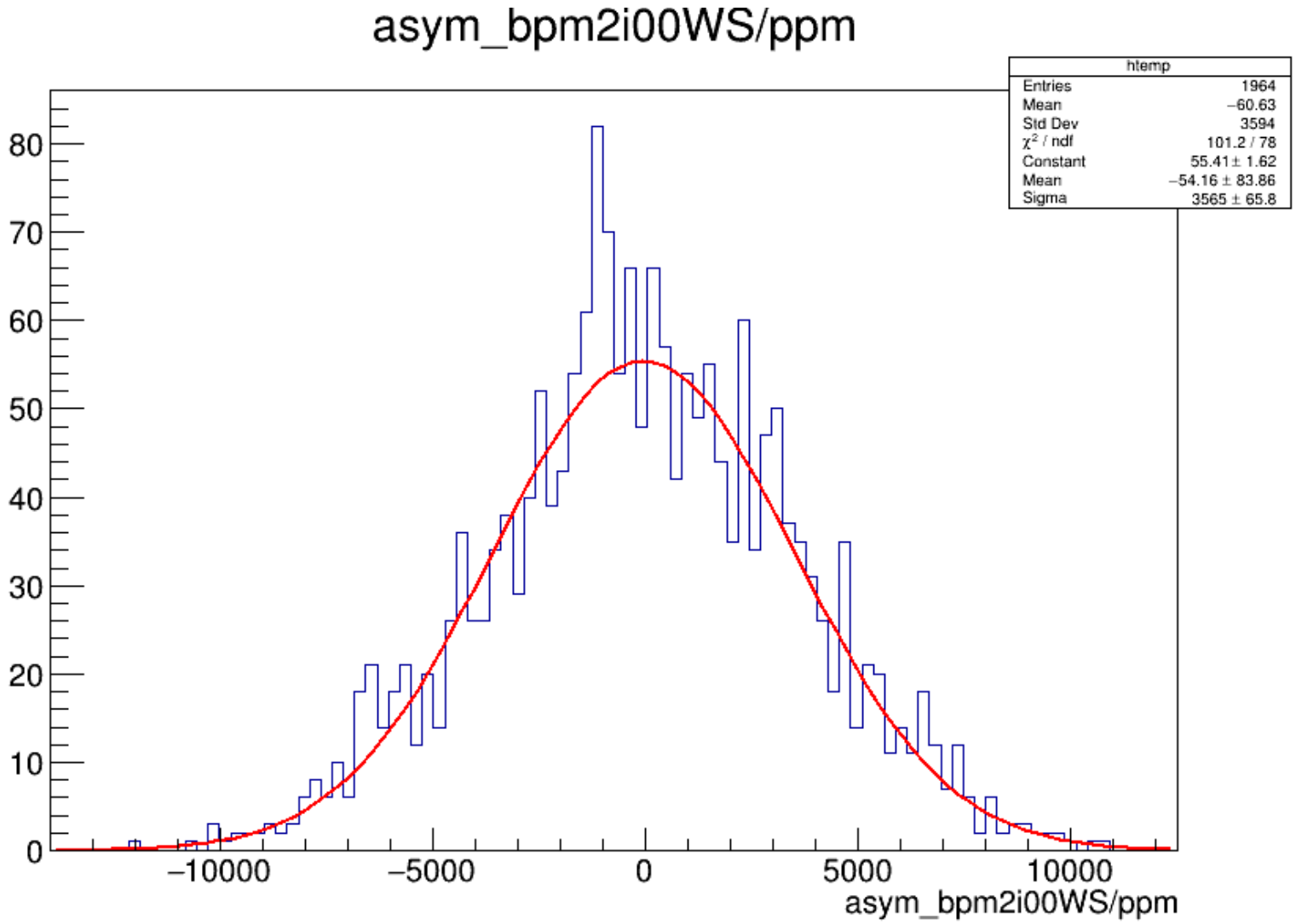
Run 20621: varied the PITA voltages V1 an V2 together (V1=V2) and measured the asymmetry. Each step corresponds to ≈60sec of data.

PITA V1,2
-9000
-7000
-5000
-3000
-1000
+1000



From the fit parameters, we obtained a PITA Voltage of -8054 counts.

PITA Voltage Check



Run 20623 (60sec)

PITA V = -8054V, $\bar{\text{Asym}} = -54 \pm 88$[ppm]

α-PosU/V

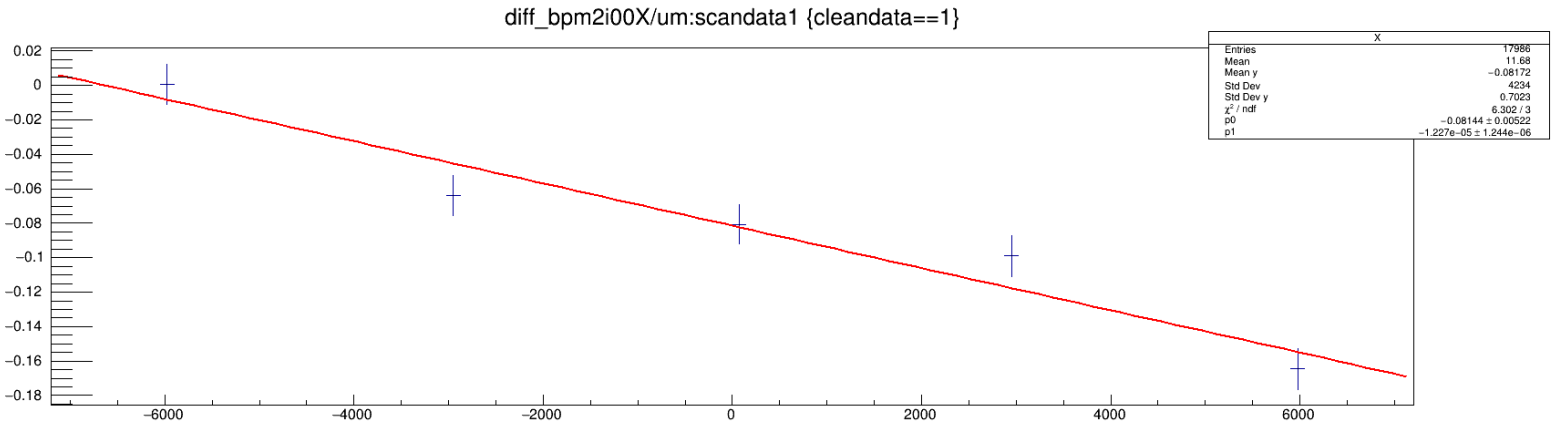
Keeping PITA V = -8054, we varied α-PosU(V) while keeping α-PosV(U) fixed at 0.

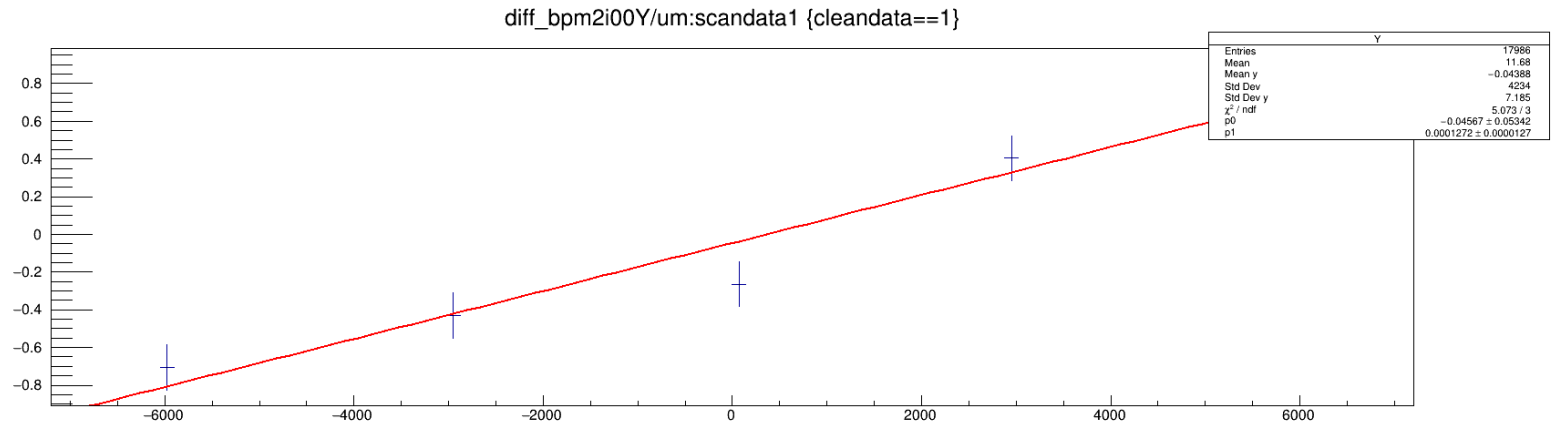
α-PosU(V)	α-PosV(U)
-6000	0
-3000	0
0	0
+3000	0
+6000	0

Each step corresponds to ≈120sec of data.

α-PosU: bpm2i00

Run 20624

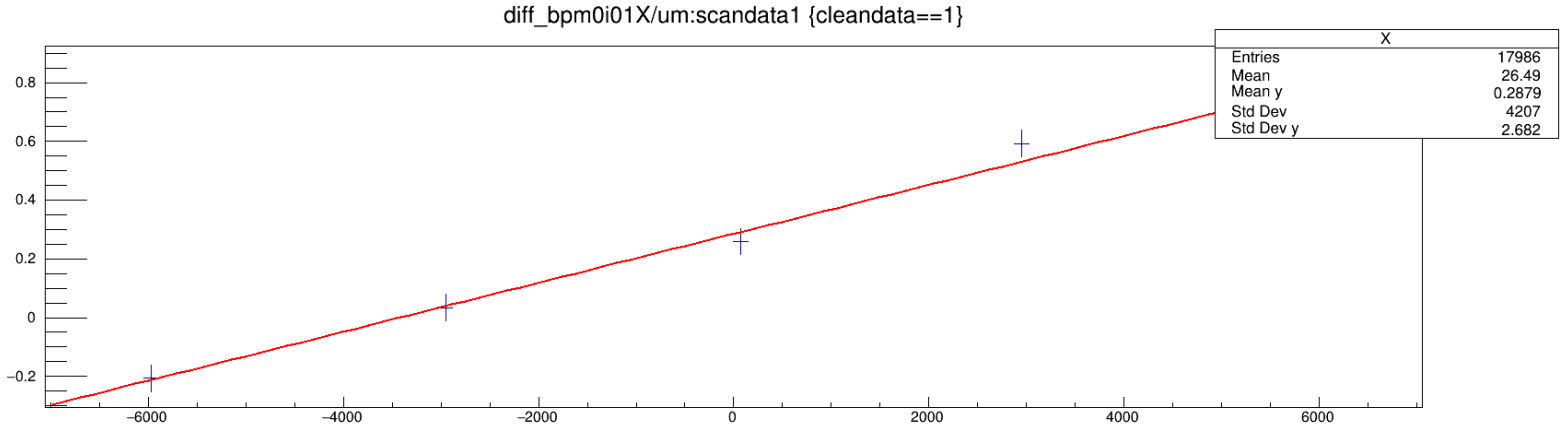


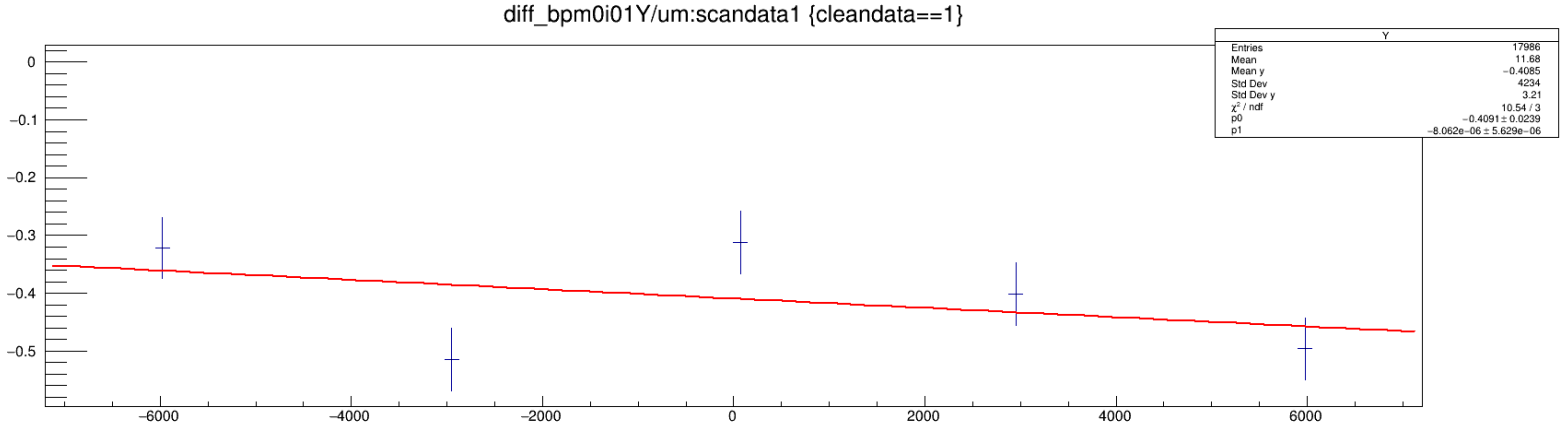


$$\boxed{\begin{aligned} V_x &= \text{-6637 [cts]} \\ V_y &= \text{-359 [cts]} \end{aligned}}$$

α-PosU: bpm0i01

Run 20624

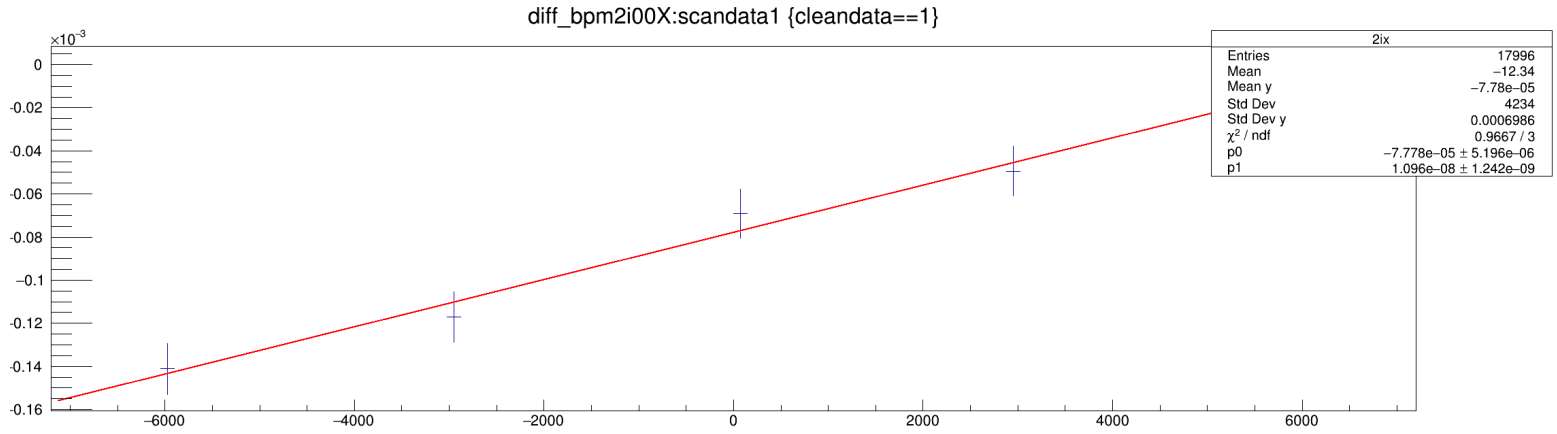


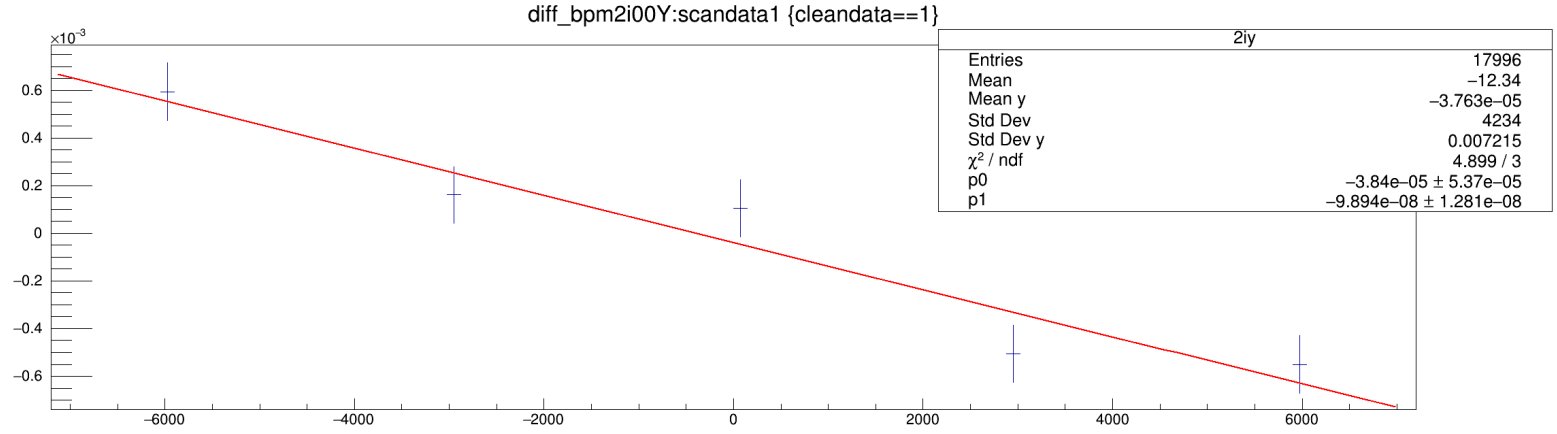


$$\boxed{\begin{aligned} V_x &= 3428 \text{ [cts]} \\ V_y &= 50744 \text{ [cts]} \end{aligned}}$$

α-PosV: bpm2i00

Run 20625

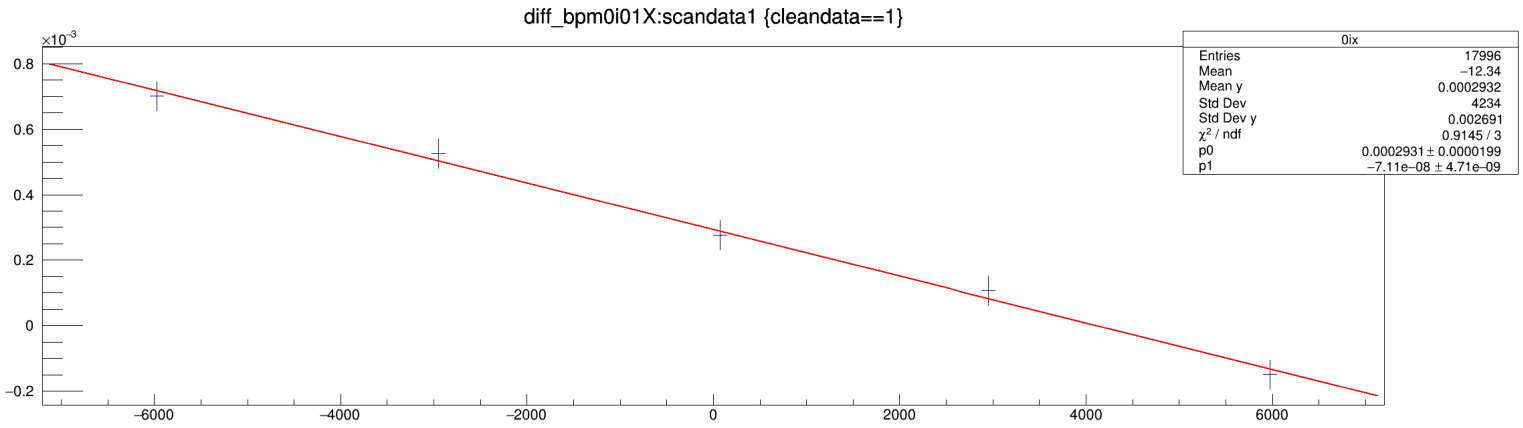


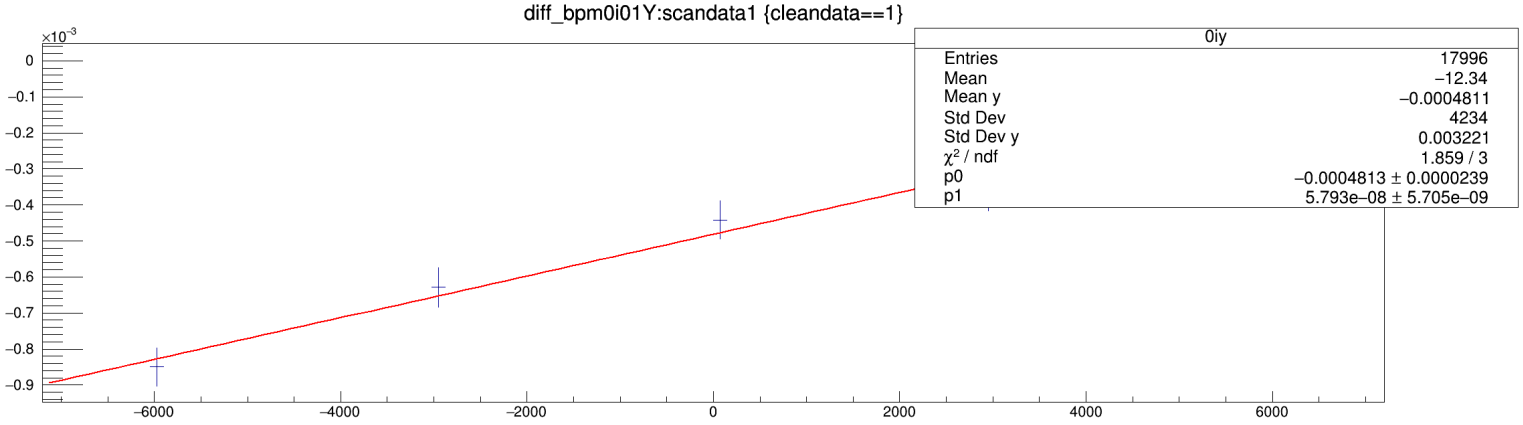


$$\boxed{\begin{aligned} V_x &= 7097 \text{ [cts]} \\ V_y &= -388 \text{ [cts]} \end{aligned}}$$

α-PosV: bpm0i01

Run 20625





$$\boxed{\begin{aligned} V_x &= \text{-4122 [cts]} \\ V_y &= 8308 \text{ [cts]} \end{aligned}}$$

RHWP Study

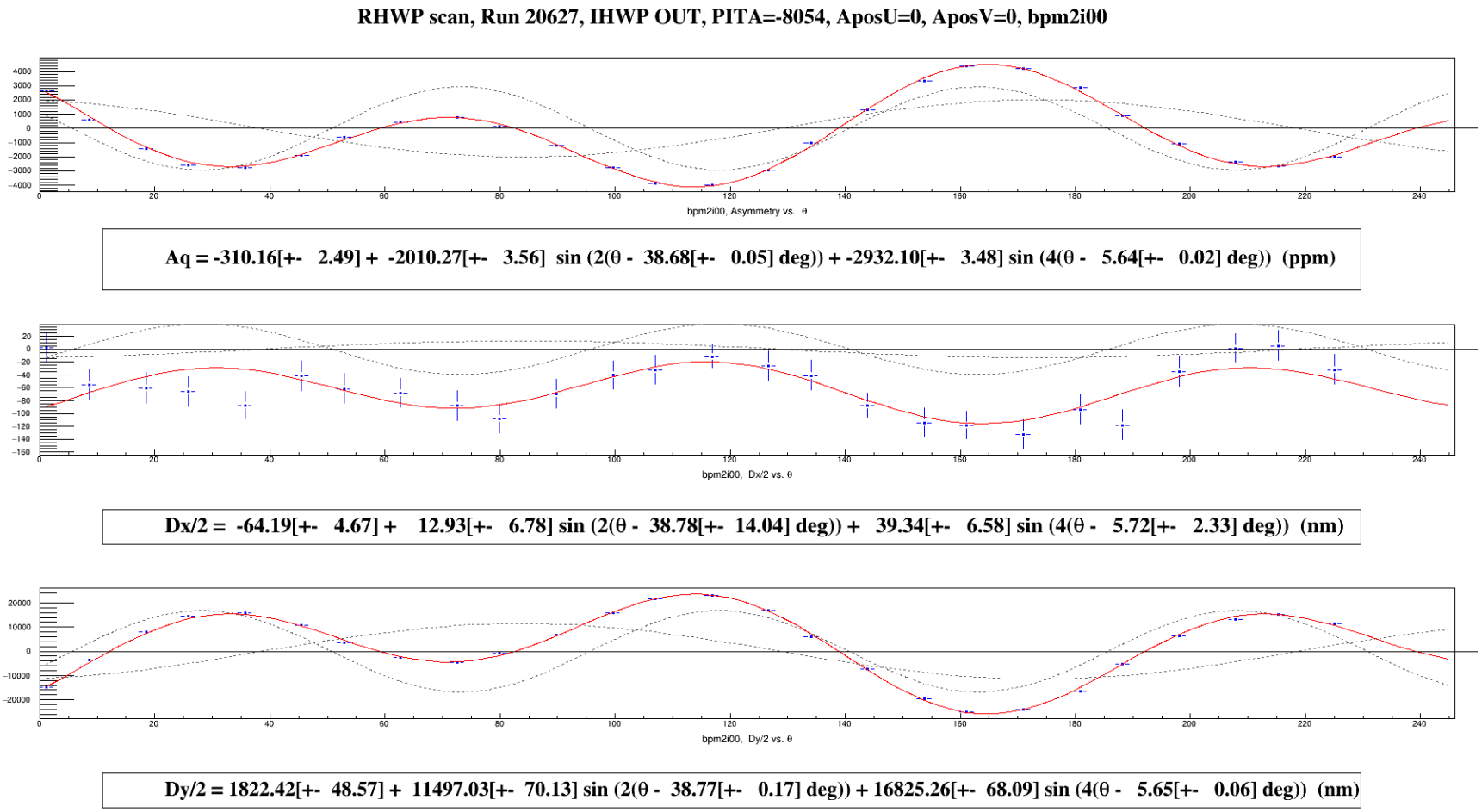
The RHWP angle was varied between [1-5000] counts in steps of 200. Each step corresponds to ≈30sec of data.

Configuration

- PITA = -8054
- α-PosU = 0
- α-PosV = 0
- IHWP = OUT (Run 20627)
- IHWP = IN (Run 20628)

IHWP = OUT (Run 20627)

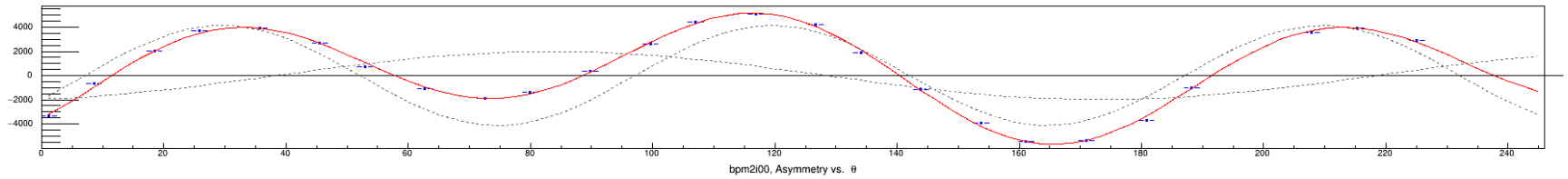
$S_1 = 5.64^\circ$, (4θ-term crosses 0)
$S_2 \approx 23^\circ$, (4θ-term is maximized)



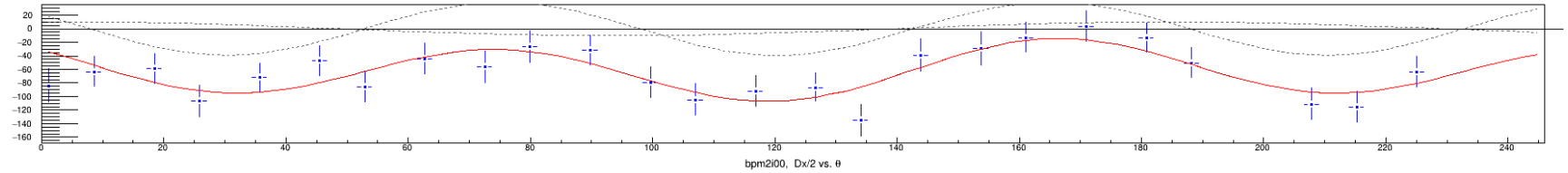
IHWP = IN (Run 20628)

$S_1 = 7.07^\circ$, (4θ-term crosses 0)
$S_2 = 29.57^\circ$, (4θ-term is maximized)

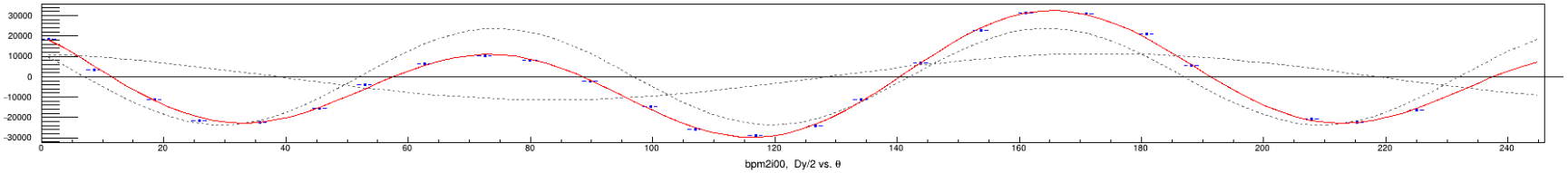
RHWP scan, Run 20628, IHWP IN, PITA=-8054, AposU=0, AposV=0, bpm2i00



Aq = 339.11[+- 2.48] + 1986.86[+- 3.65] sin (2(θ - 38.30[+- 0.05] deg)) + 4126.26[+- 3.55] sin (4(θ - 7.07[+- 0.01] deg)) (ppm)



Dx/2 = -61.66[+- 4.81] + -10.09[+- 6.75] sin (2(θ - 48.97[+- 19.34] deg)) + -38.96[+- 6.76] sin (4(θ - 7.80[+- 2.45] deg)) (nm)



Dy/2 = -1989.65[+- 48.57] + -11316.84[+- 69.64] sin (2(θ - 38.56[+- 0.17] deg)) + -23661.40[+- 68.93] sin (4(θ - 6.98[+- 0.04] deg)) (nm)